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| SENSITIVE |

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MILITARY SPECIFICATION

MIXING PLANT, ASPHALT, DRUM TYPE, PORTABLE, ELECTRIC MOTOR DRIVEN, 120 TONS PER HOUR CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric motor driven, portable, asphalt mixing plant having a mixing capacity of not less than 120 tons per hour.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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TT-V-119 - Varnish, Spar, Phenolic-Resin.
VV-F-800 - Fuel Oil, Diesel.
VV-L-751 - Lubricating Oil: Chain, Wire-Rope, Exposed Gear.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproof, Flexible.
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool and Tool Accessories For Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant) Packaging of.
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-V-13811 - Varnish, Waterproofing, Electrical Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-G-28554 - Generator Sets, Mobile Electric Power and Supplemental Equipment: Packaging of.
MIL-C-52950 - Crates, Wood, Open and Covered.
MIL-P-53030 - Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-1186 - Cushioning Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.

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2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

CODE OF FEDERAL REGULATIONS (CFR)

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, CFR, Part 1910 - Occupational Safety and Health Standards Health Standards, and Part 1926 - Safety and Health Regulations for Construction.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, CFR, Part 60 - Standards of Performance for New Stationary Sources.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

DEPARTMENT OF TRANSPORTATION (DOT)

DOT - Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B15.1 - Mechanical Power - Transmission Apparatus.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 27 - Classification of Fireclay and High-Alumina Refractory Brick (DOD adopted).
ASTM D 3951 - Commercial Packaging (DOD adopted).

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems (DOD adopted).
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies (DOD adopted).
NEMA ICS 6 - Enclosures for Industrial Controls and Systems (DOD adopted).
NEMA MG 1 - Motors and Generators (DOD adopted).
NEMA WC 3 - Rubber Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.
NEMA WC 7 - Cross-Linked-Thermosetting Polyethylene-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy (DOD adopted).

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code (DOD adopted).

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

NATIONAL LUBRICATING GREASE INSTITUTE (NLGI)

NLGI - Lubricating Grease Guide.

(Application for copies should be addressed to the National Lubricating Grease Institute, 4635 Wyandotte Street, Kansas City, MO 64112.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings (DOD adopted).
SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30 - 1000 MHz) (DOD adopted).
SAE J700 - Upper Coupler Kingpin - Commercial Trailers and Semitrailers.
SAE J702 - Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer, Recommended Pracitice (DOD adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES INC. (UL)

UL - 486E - Equipment Wiring Terminals for Use with Aluminum and/or Copper Conductors (DOD adopted).

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The asphalt mixing plant, hereinafter called the asphalt plant, shall be an electric motor driven, portable drum type. The asphalt plant shall be comprised of towable main component units, as specified herein, with the supplemental components, units, and accessories to be transported by truck or trailer. The asphalt plant shall be capable of mixing not less than 120 tons per hour of high quality asphaltic mix and reclaimed asphaltic material. The asphalt plant shall include, but not be limited to the following basic units:

a. One four-bin cold-aggregate feed unit of hopper and frame construction with gathering conveyor used for blending and feeding predetermined amounts of cold-aggregate to the cold-aggregate conveyor.

b. One cold-aggregate conveyor system to receive cold-aggregate from the feeder bin gathering conveyor and discharge into the drum mixer dryer.

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c. One drum mixer of the single rotating type with burner and blower to heat and dry the aggregate and mix with asphalt.

d. One wet or dry dust collection system for air pollution control with fines recovery capability.

e. One totally enclosed conveyor for conveying hot asphalt mix from the drum mixer to the surge bin.

f. One surge bin system to receive the hot asphalt mix, with power operated dual discharge gates for truck loading.

g. One complete set of insulated piping and accessories necessary for interconnecting all the component units.

h. One wheel mounted combination asphalt and fuel oil storage tank with 30,000 gallons capacity, with pumps and metering system.

i. Plant control trailer complete with motor controllers, circuit breakers, switches, relays, transformers, and auxiliary equipment, properly wired and operable as a central control for the asphalt plant.

j. Portable power and control cables of the proper size and number necessary to provide power and control circuits from the control centers for operation of the asphalt plant.

k. When specified (see 3.10.4 and 6.2), a skid mounted diesel engine driven generator set.

The above units shall be semi-trailer mounted and may be grouped at the discretion of the contractor as long as the asphalt plant can be interconnected mechanically and electrically, and operated to rated capacity.

3.1.1 Recycling capability. The asphalt plant shall be equipped with all necessary components and accessories to recycle reclaimed asphaltic material.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Standard commercial product. The asphalt plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the asphalt plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.4 Material. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be inter-preted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically, and all moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel, shall be enclosed or guarded. Protective devices shall not impair the operating functions. All asphalt plant components shall conform to the applicable safety and health standards and regulations of DoL OSHA, Title 29, CFR, Part 1910 and Part 1926. Each trailer shall conform to all applicable requirements of the DOT Motor Carrier Safety Regulations. Reflectors are required. Steel walkways and ladders shall be provided where necessary for inspection, servicing, and maintenance. Walkways shall be provided with safety tread and handrails.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Maintainability. The asphalt plant design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional, general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the asphalt plant.

3.8 Servicing and adjusting. Prior to acceptance of the asphalt plant by the Government, the contractor shall service and adjust the asphalt plant for immediate operational use. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

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The asphalt plant shall be conspicuously tagged to identify the lubricants and their temperature range.

3.9 Vehicular component units. The semi-trailer component units shall be as specified herein. Exclusive of the running gear, the ground clearance of each unit shall be not less than 10 inches.

3.9.1 Intervehicular connections.

3.9.1.1 Truck-tractor to semi-trailer. The semi-trailer shall be furnished with airbrake standard commercial couplings, for connecting the truck-tractor airbrake system to the semi-trailer system. The coupling shall be installed on the front of the semi-trailer, symmetrical about the centerline, in accordance with SAE J702. Coupling clearance shall permit connection of the truck-tractor hoses to the semi-trailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings shall be protected with spring covers.

3.9.2 Chassis frame. The framing members of the component units shall support the weights and maintain alignment under all conditions of transport and of asphalt plant operation as specified herein.

3.9.3 Semi-trailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a lower fifth-wheel, 36 inches in diameter, and the entire width of the gooseneck. The fifth-wheel plate and kingpin shall be compatible with a fifth-wheel height of 48 inches $\pm$1.0 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include not less than four vertical gussets and shall be provided with holes for water drainage. A kingpin with a throat diameter of 2 inches and conforming to SAE J700 shall be installed not less than 24 inches nor greater than 36 inches from the front of the semi-trailer. The center of the kingpin shall be within 0.125 inch of the longitudinal centerline of the semi-trailer. The swing clearance rearward from the kingpin to any part of the semi-trailer below the upper fifth-wheel plate shall be not less than 74 inches.

3.9.4 Landing jacks. Landing jacks shall be provided on the semi-trailers for raising, leveling and supporting each vehicular component unit. The landing jacks shall be hand or power operated hydraulic units, and shall be permanently installed on the vehicular component unit. The landing jacks shall be of such strength, design, and location that coupling and uncoupling the trailer, with a towing vehicle, shall not cause excess stress. Operating jacks shall be furnished to raise each vehicular component unit into position for asphalt plant operation. Operating jacks shall be of such strength, design, and location that operation of the asphalt plant under load will not cause excess or uneven stresses. All jacks shall be equipped with square metal mud pads having a ground bearing area of not less than 100 square inches.

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3.9.5 Running gear. The axle, or the forward axle on tandem axles, shall be in alignment with the longitudinal centerline of the vehicular component unit so that the two diagonal measurements, from the kingpin to the outer center of each hub, do not differ greater than 0.125 inch. The rear axle of tandem axle assemblies shall be in alignment with the front tandem axle so That the parallel measurements between hub centers on each side do not differ greater than 0.125 inch. Stub axles or through axles may be furnished. The length of axle tubes of through axles may be increased during manufacture, providing the mounting pads are increased not less than an equal distance to insure that the structural rigidity is equal to that of the axles. Axle assembly for the component units shall be either of the single or tandem axle type, as applicable for load distribution and weight. All units shall be equipped with suitable size wheels and tires, in accordance with the recommendations of the TRA Yearbook, to withstand the maximum loads, stresses, travel speeds, and with adequate road clearance. Each single axle shall be designated for a load of not greater than 18,000 pounds (lbs) and shall be compatible with the brake system in accordance with the loadings.

3.9.6 Brake system. Each wheel component unit shall be equipped with a service and emergency breakaway, waterproof, all air or air over hydraulic brake system complying with all applicable construction and performance requirements of the DOT Motor Carrier Safety Regulations. The brake system includes service and emergency line filters, an air reservoir, and an emergency relay valve. All brake system components shall be located, installed, and protected so that damage from ground obstructions and debris is prevented and that maximum efficiency and ease of maintenance is obtained.

3.10 Electric equipment.

3.10.1 Electric motors. All electric motors shall be single speed, induction type, and connected for 460 Volts alternating current (Vac), 3-phase, 60 Hertz (Hz) operation. All motors shall be totally enclosed, nonventilated, or totally enclosed fan-cooled machines conforming to NEMA MG 1. Motors shall be suitable for the respective applications in an ambient temperature of 125 degrees Fahrenheit (°F) [51.7 degrees Celsius (°C)]. All motors, 25 horsepower (hp) and larger, shall be arranged for sequence starting from nonreversing reduced voltage or current motor starters. All motors less than 25 hp may be across-the-line start with overload protection controlled from the motor control center (see 3.18).

3.10.2 Motor controls. The motor controls shall be a combination motor control unit, rated for 460 Vac, 3-phase, 60 Hz, 3-wire, and shall conform to NEMA ICS 1 and ICS 2. Motor controls shall be enclosed in a NEMA ICS 6 Type 3S enclosure. Provision shall be made for remote control stations (start-stop pushbuttons) where required. All motor controller, components, and accessories, shall be suitable for continuous operation under conditions as follows:

a. At ambient temperatures between +32°F to +125°F (0°C to 51.7°C) and at any possible relative humidity.

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b. Exposure to severe dust conditions.

c. Exposure to salt air.

d. At any altitude from mean sea level to 5,000 feet (ft) above mean sea level.

The motor controls shall have a manually operated main circuit breaker on the incoming power supply to the control center. The main circuit breaker may be either integral with the control center, or in a separate enclosure and shall have overload, ground fault, and short circuit protection. The main circuit breaker shall have a shunt trip, with remote trip pushbutton(s) located at the operator position(s), for emergency trip operation. The termination for the incoming power supply connectors to the main circuit breaker shall have screw pressure type terminal for not less than three sets of power supply cables. One set of terminal connectors (one per phase) shall be suitable for terminating conductors from number 4 American Wire Gage (AWG) to number 1 AWG; the other terminals shall be suitable for terminating conductors from 300,000 circular mils to 600,000 circular mils. Connectors shall be in accordance with UL486E. The asphalt plant control center shall have an instruction plate (see 3.25.1) permanently affixed in a conspicuous location. In addition to instructions for connecting and operating the asphalt plant, the instruction plate shall show the power supply requirement of the asphalt plant in bold type, as follows:

POWER SUPPLY REQUIRED
460 VOLTS, 3-PHASE, 60 HERTZ
INPUT CAPACITY, MAXIMUM _____ KVA (_____ KW)
OPERATING LOAD (AT RATED CAPACITY) _____ KVA (_____ KW)

3.10.2.1 Motor controllers. Motor controllers (starter) shall be alternating current, general purpose, class A, magnetic controllers for single speed, nonreversing, nonjogging, squirrel-cage induction motors, conforming to NEMA ICS 2. Controllers for motors 15 hp and larger shall be reduced voltage or current motor controllers. Controllers for motors less than 15 hp may be full-voltage (across-the-line) controllers. For consistency in asphalt plant operation, the controllers shall be interlocked to provide sequence starting, with the larger motors to be started first, in order to reduce the required power input capacity to a minimum.

3.10.3 Power and control cables. All power, control and grounding cables, fittings, and accessories required to completely connect the asphalt plant for operation shall be furnished in the length required for each circuit. In addition, three sets of power cables (one three-conductor cable or three one-conductor cables each), not less than 100 feet in length, shall be furnished to connect the motor control center to the incoming power supply source. One set of power supply cables shall have adequate load capacity to supply the standby loads, such as the hot oil heating system and asphalt storage transfer pumps, but the individual conductors shall be not smaller

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than number 4 AWG. The other sets of power supply cables shall have adequate load capacity to supply the maximum power input required to operate the complete asphalt plant, but all the individual conductors shall be not smaller than 400,000 circular mils copper. All power and control cables, for the power supply and for interconnecting the asphalt plant, shall be heavy-duty portable cables conforming to NEMA WC 3 or NEMA WC 7 and suitable for direct burial. All portable cables shall be rated for conductor operating temperature of not less than 167°F (75°C). The load on any conductor shall not be greater than the allowable ampacities of NFPA 70 for direct burial conductors. The grounding system for the complete asphalt plant shall include the grounding conductors, fittings, ground rods, and clamps. The grounding conductors shall be of suitable size and type for the application. Grounding shall conform to the wiring practices of NFPA 70.

3.10.4 Skid mounted diesel engine driven generator set. When specified (see 3.1 and 6.2), the manufacturer shall provide a type II, style A, class 1 skid mounted diesel engine driven generator set (referred hereinafter as the generator set) conforming to MIL-G-19826. The generator set shall be capable of providing all power requirements at the voltages specified in 3.10 to operate all electrical motors and accessories of the asphalt plant under the loads specified in 3.20.

3.10.4.1 Electromagnetic interference suppression. When specified (see 6.2), the generator set shall meet the electromagnetic interference suppression limits of SAE J551.

3.11 Drive components.

3.11.1 V-belt drives. V-belt drives shall shall be in accordance with ANSI B15.1 and shall have provisions for adjustment.

3.11.2 Chains, gears, and bearings. All chains, gears, and bearings shall have provisions for lubrication. Bearings shall be lubricated anti-friction type with dust shield provisions.

3.12 Cold-aggregate feeder(s). The cold-aggregate feeder(s) shall be of hopper and frame construction and shall consist of a four-bin aggregate hopper, with a total capacity of not less than 100 tons, belt-conveyor feeder(s), all supported by a rigid steel frame and mounted on a pneumatic-tired semi-trailer chassis. The cold-aggregate feeder(s) shall discharge the materials into the cold-aggregate conveyor. Bin vibrators shall be provided for each compartment.

3.12.1 Hoppers. The four-bin aggregate hopper shall be of welded steel construction. The bin side plates and divider plates shall have a wide opening for receiving the amount of aggregate being dumped into each hopper. Hopper and bin extensions shall be provided for each hopper. Bins and bin extensions shall be provided with reinforcing to prevent damage to the edges while bins are being charged. Each bin shall be provided with an individual,

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adjustable, control gate, or by other suitable means, to regulate the flow of aggregate. The gate shall be provided with an adjustment lever and position locking device when control gate is used for regulating aggregate flow.

3.12.2 Belt-conveyor feeder(s). The belt-conveyor feeder(s) shall consist of a suitable steel frame, conveyor belt, head-end (drive) pulley, foot-end pulley, carrier rolls, and return rolls. Straight carrier rolls shall be mounted under the belt to support the aggregate as it is discharged from the aggregate hoppers. Steel dustpans shall prevent dust and aggregate from falling onto the backside of the conveyor belt. Steel side plates and rubber flashing shall be provided to prevent spillage of aggregate from the sides as it is discharged from the hopper. Pulleys and rollers shall be mounted on steel shafts and supported by anti-friction or bronze bearings with dustproof seals. Self-alining bearings for head and tail end pulley shafts shall be furnished and shall be provided with labyrinth seals. The belt-conveyor feeder(s) foot-end section shall be provided with a takeup adjustment mechanism to obtain belt tension.

3.13 Cold-aggregate conveyor. Cold-aggregate conveyor shall be of the belt conveyor drive type of not less than 24 inches in width and the manufacturer's standard length, with masts or arms for elevating the frame to an angle of not less than 20 degrees (°). The conveyor frame shall be raised either hydraulically or by mechanical means. The chassis shall be constructed so that the entire unit can be towed with a drawbar.

3.13.1 Scalping screen. When specified (see 6.2), a single deck scalping screen mounted over the tail section of the aggregate feed conveyor for scalping out oversize material and other debris shall be provided. The unit shall include drive, motor, and screen cloth.

3.14 Drum mixer unit. The drum mixer unit shall be semi-trailer mounted and shall include, but not be limited to, the drum mixer, aggregate feed system, burner and fuel system, fuel asphalt feed system, hot mix discharge system, drive mechanism, and, dust and fines collection system. The drum mixer unit shall, in addition, be capable of mixing recycled reclaimed asphaltic material as specified in 3.1.

3.14.1 Drum mixer. The drum mixer shall be a single cylinder constructed of heavy-gage steel plate capable of withstanding the abrasion wear and heat distortion at peak production for long usage without replacement. The drum shall be provided with replaceable lifting flights which shall lift and drop the aggregate in a thin veil through the hot gases produced by the burner and then mix with hot asphalt. Suitable means shall be provided to discharge the hot mix into the hot-mix conveyor. The drum shall be fitted with steel supporting tires attached to the outer circumference. Each tire shall revolve on not less than two adjustable trunnions made of hardened cast steel. Means shall be provided for controlling the longitudinal movement of the drum along the longitudinal axis of the drum in both the uphill and downhill direction. The drum shall be driven by a positive or friction drive arrangement. All bearings used in the driving mechanism shall be of the antifriction type.

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3.14.2 Burner and burner fuel system. The burner and burner fuel system shall include burner, forced draft blower, fuel piping, fittings, valves, fuel pump, strainers, sediment and moisture bowl, relief valve, pressure gages, and other items which may be required to provide a complete operating burner and burner fuel system.

3.14.2.1 Burner. The mixer burner shall be low pressure, air atomizing and oil burning. The burner shall meet the performance requirements specified herein, when operated with fuel oil conforming to VV-F-800, grade DF-A, DF-1, or DF-2, and jet fuel conforming to MIL-T-5624, grade JP-4 or JP-5. The burner shall have the capacity of drying the wet aggregates to meet the specified production requirement, and shall operate in conjunction with the forced draft blower for efficient combustion.

3.14.2.1.1 Forced draft blower. A centrifugal type blower shall be provided to supply air for atomization to the burner. The forced draft blower shall have a centrifugal type wheel, steel shaft, housing, and drive. The forced draft blower shall have a capacity to provide a sufficient quantity of air to the burner to assure efficient operation of the burner at maximum burner capacity. The forced draft blower shall be provided with antifriction type bearings, and the intake shall be provided with an expanded metal or wire mesh screen.

3.14.2.2 Burner fuel system.

3.14.2.2.1 Fuel piping and pipe fittings. Fuel pipe shall be black iron and pipe fittings shall be malleable iron. Unions shall be installed, where required, for disassembly of piping, valves, and accessories. On the discharge line of the fuel oil pump, a tee-off line with stop valve shall be provided for refilling the fuel tanks of auxiliary generator sets.

3.14.2.2.2 Fuel control device. A control device shall be installed in the fuel supply system to the burner in a location accessible to the operator. The control device shall provide an infinitely variable range of adjustment and shall maintain a steady flow of fuel at any setting. The control device shall include a calibrated dial and indicating pointer to facilitate burner operating procedures. A gate type shutoff valve shall be installed in the fuel supply piping between the control device and the fuel pump. The gate valve shall be of the same size as the piping in which it is installed. A relief valve shall be provided and installed in the fuel supply line to the burner. The relief valve shall have a capacity not less than the capacity of the fuel pump. A fuel tank return line shall be connected to the discharge of the relief valve. All low points in the fuel oil system shall be provided with drains to permit complete drainage of the piping.

3.14.2.2.3 Fuel pump. The fuel pump shall be of the positive displacement type for pumping the required fuels to supply the heat demand. This capacity shall be developed when pumping against a discharge pressure not less than the combined friction loss created by the burner and fuel supply system. The pump shall be self-priming, with an electric motor drive, and shall be permanently mounted on the mixer frame.

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3.14.2.2.4 Strainer. An oil strainer shall be provided and installed in the fuel supply line in such a manner as to permit easy accessibility for cleaning. The strainer shall have the capacity to remove all foreign matter which might foul the burner.

3.14.2.2.5 Advance temperature detector. An advance temperature detecting device shall be provided for compensating the sudden surge of fuel temperature relationship demand in the mixer when charged with large load of aggregate with higher or lower than normal moisture content. The device shall be capable of overriding the set fuel control until temperatures are stabilized.

3.14.2.2.6 Fuel pressure gage. A fuel oil pressure gage shall be installed in the burner fuel supply system. The gage shall operate with the fuel oils specified for the fuel pump, shall have a heavy-gage metal case and an indicating dial with a range of 0.0 to 100 pounds per square inch gage (psig), and shall be provided with a shutoff cock or valve. The dial shall be graduated in increments of not less than 5 pounds per square inch (psi).

3.14.2.2.7 Combustion chamber and ignition cone. The combustion chamber shall consist of a steel cylinder lined with firebrick conforming to ASTM C 27, refractory, or plastic refractory with equal rating. The ignition cone shall consist of either a steel refractory lined cone or a stainless steel refractory cone.

3.14.2.2.8 Temperature indicator. A temperature indicator shall be provided to indicate the temperature of the mix as it is discharged from the drum mixer. The indicator shall have a range of 100°F to 500°F (37.8°C to 260°C).

3.14.3 Level indicator. A level indicator shall be provided and shall be mounted on the drum mixer assembly in such a manner as to indicate the angle of the longitudinal axis of the drum mixer above the horizontal. The indicator shall include a graduated scale in increments not greater than 1.0° and with a range of 0.0° to 10.0°. The indicator shall be mounted in a location where it will be visible and legible to the operator.

3.14.4 Driving mechanism. All necessary gearboxes, shafting, motor, chains, bearings, and any components required for a complete operating drive system shall be provided.

3.14.4.1 Gearboxes. The main drive shall be provided with a gearbox to the power source. Gears shall run in an oil bath. Gearboxes shall be provided with oil seals to prevent loss of oil and entry of foreign material.

3.14.5 Discharge conveyor and discharge chute. Means shall be provided at the end of the mixer drum to discharge the hot mix into a discharge chute which will, in turn, direct the mix into the boot of the hot-mix conveyor. The conveyor discharge shall be of such a height to eliminate the need for a hot-aggregate conveyor pit.

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3.14.6 Fines and dust collection system(s). The fines and dust control system(s) shall be mounted on the mixer drum semi-trailer or on a separate semi-trailer. The system(s) shall include all the necessary components for a complete system(s) such as motor, exhaust fan, steel stack, conveyor, collector(s), duct work, instruments, accessories, and other components for the operation of the unit. The system(s) shall meet the particulate emissions requirements of EPA, Title 40, CFR, Part 60.

3.15 Surge bin (silo) system. The surge bin system shall be complete with batcher, surge bin, loading and unloading panel, and necessary air and electrical components. The bin or cone shall be constructed of not less than 0.25 inch steel plate and shall be insulated. Capacity shall be not less than 35 tons using material weighing 100 pounds per cubic foot. Dual power operated discharge gages approximately 15 inches wide by 54 inches long shall be provided. Driveway truck clearance shall be not less than 10.5 feet wide by 12.0 feet high. High and low level indicators for the bin and a weighing system for weighing the mix being loaded into the trucks shall be provided.

3.16 Fuel oil and asphalt storage tank. A semi-trailer mounted combination fuel oil and asphalt storage tank, with finned heating coils submerged in the asphalt tank shall be provided. The tank shall have two compartments with double wall insulated bulkhead to provide capacities for not less than 18,000 gallons of liquid asphalt and 12,000 gallons of fuel oil.

3.16.1 Tank construction. The tank shall be constructed of not less than 0.25 inch thick steel plate with 3.50 inches of fiberglass insulation and covered with an outside steel jacket with a thickness of 0.06 inch. The tank shall be reinforced with adequate angle rings spaced at intervals and channel stiffeners on the tank heads. The tank shall be supported on saddles spaced at intervals and two full length channels or beams. Each tank compartment shall be provided with an 18-inch manhole with locking cover and internal and external ladders. Level indicators for each tank compartment shall be provided.

3.16.2 Heating system. The tank shall be equipped with a complete heating system for heating the liquid asphalt and fuel oil. The system shall include a 1,500,000 British thermal units per hour automatic type burner, fire tube heat exchanger to heat the asphalt and a finned tube heat exchanger for the hot oil system. An expansion tank with a capacity of 120 gallons shall be furnished mounted and piped. All operating components, accessories and controls shall be provided and mounted on the semi-trailer gooseneck. A pipe rack shall be provided on one side of the semi-trailer for transporting the jacketed asphalt piping.

3.17 Piping and accessories. All piping and accessories, which are required for interconnecting of component units to form a complete operating asphalt plant, shall be furnished. Where sectionalization is required on the hot oil asphalt lines for assembling and disassembling of the asphalt plant, flange connections with fiber gaskets shall be installed. Wire braided

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flexible hoses shall be provided for the hot oil and asphalt lines. All piping, fittings, valves, and other accessories to make up a complete asphalt plant shall be of steel, iron body, or malleable iron. Burner fuel piping shall be capable of withstanding 100 psi hydrostatic test pressure; hot oil piping, including jackets, 100 psi; and asphalt piping, 50 psi.

3.18 Plant control trailer. The asphalt plant control trailer shall be semi-trailer mounted and shall include all controls, such as weighing scales, thermometers, gages, indicating light signals, levers, drier burner controls, and other accessories for the operation of the drum mixer asphalt plant shall be located in the control trailer. Unless otherwise specified (see 6.2), the asphalt plant control system shall be the manufacturer's standard for the asphalt plant offered.

3.19 Automatic mixing control system. When specified (see 6.2), an automatic mixing control system shall be furnished. Provisions shall be made for switching the system to manual operation.

3.20 Performance.

3.20.1 Capacity. The asphalt plant shall be capable of producing not less than 120 tons per hour at a mix discharge temperature of 275°F (135°C) with uniformly graded nonsticky material, either new or reclaimed or both, having a free moisture content of 5.0 percent.

3.20.2 Transportability. Each wheeled component, when towed, shall be capable of travel over smooth hard-surfaced primary highways at speeds up to 20 miles per hour (mph), travel over unimproved roads and hilly cross-country terrain, and negotiate a 90-degree turn without weaving.

3.20.2.1 Braking ability. The combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance with DOT Motor Carrier Safety Regulations requirements. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.21 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated as specified in section 5.

3.22 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections shall be coated with varnish conforming to MIL-V-173, except that:

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a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.

b. Current-carrying contact surfaces, such as relay contact points shall not be coated.

3.23 Lifting and tiedown attachments. The asphalt plant shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A non-ferrous transportation plate shall be provided and mechanically attached to the asphalt plant. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall indicate which attachments are intended for the tie down of the asphalt plant on the carrier when shipped. The design of lifting devices shall exclude the use of spreader bars.

3.24 Cleaning, treatment, and painting. Surfaces painted shall be cleaned, treated, and painted as specified herein. Unless otherwise specified (see 6.2), the color of the finish coat shall be green, conforming to FED-STD-595 color number 14064. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, greases, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to the manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.25 Data plates.

3.25.1 Instruction plates. The asphalt plant shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 0.125 inch diameter.

3.25.2 Nameplate. A nameplate will be furnished by the contracting officer for each component unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose, and securely affix it to

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each asphalt plant in a conspicuous place with nonferrous screws, bolts or rivets not less than 0.125 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

3.26 Special maintenance tools. Such special tools as are required for field maintenance and operation of the equipment shall be packed in the toolbox and shipped with the equipment. Special tools are defined as those not normally included in a mechanic's tool kit, but which are designed specifically for a certain purpose on a specific piece of equipment. Means shall be provided to carry those tools and accessories required to operate the asphalt plant. Toolboxes, when furnished to hold tools and accessories, shall be made of sheet steel 0.0747 inch (U.S. revised standard gage number 14) nominal thickness.

3.27 Workmanship.

3.27.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made by controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.27.2 Bolted connections. Boltholes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.27.3 Riveted connections. Rivet holes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.27.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.27.5 Castings. All castings shall be sound and free from patching misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practic to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete asphalt plant when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Each asphalt plant produced shall receive the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each asphalt plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with

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specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall be tested as specified in 4.4.1 through 4.4.5. Each production unit shall be tested as specified in 4.4.1 and 4.4.2. Failure to pass any test shall constitute cause for rejection.

4.4.1 Test conditions. Prior to being tested, the asphalt plant shall be lubricated. Oils shall be those designated for use in the ambient temperature at which the tests are being performed. During all asphalt plant tests, the motors shall be run at normal operating speeds. Where individual components are tested with the use of power sources other than the asphalt plant power supply, the component will be operated at the normal asphalt plant operating speeds. When the generator set is required (see 3.1), the asphalt plant shall be disconnected from the normal power supply and connected to the generator set. The generator set shall supply all the necessary electrical power to operate all components and accessories of the asphalt plant at normal operating speed. As services and adjustments are performed, the asphalt plant shall be checked for ease of maintenance.

4.4.2 Compatibility. Assemble and operate all the component units. The test may be conducted with load or without loads for not less than 0.5 hour, provided that operating with or without loads will cause no damage to any component unit or to the asphalt plant. All component units shall assemble and operate compatibly without modifications. When the generator set is required (see 3.1), the asphalt plant shall be disconnected from the normal power supply and connected to the generator set. The compatibility test shall then be repeated, using the generator set as the sole power supply for the asphalt plant. The piping, hose, valves, fittings, and accessories shall be assembled, and matchmarks shall be painted on each of the piping, fittings, valves, and accessories, before disassembled for shipment, for the purpose of easy identification when reassembled in the field.

4.4.3 Capacity test. The asphalt plant shall be tested for the capability of producing not less than 120 tons per hour of drying and mixing new aggregate or recycled asphaltic mix having 5.0 percent free moisture content.

4.4.4 Vehicular component units transportability test. All component units equipped with axles and rubber tired wheels shall be towed by a wheeled tractor or truck tractor at a speed not greater than 20 mph, over smooth hard surfaced roads for a distance of not less 5.0 miles, then towed over unimproved roads for not less than 0.5 mile. During or after the test on each trailer, the trailing, turning, and braking ability shall be observed.

4.4.5 Lifting and tiedown attachment tests. When lifting and tiedown attachments are required, the asphalt plant shall be tested to conform to 3.23.

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4.4.6 Electromagnetic interference suppression. When specified (see 6.2), the generator set shall be tested in accordance with SAE J551 to meet the requirements of 3.10.4.1.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The generator set, when provided (see 3.1), shall be preserved as specified in MIL-G-28554. The asphalt plant shall be preserved as specified in the following paragraphs.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts and assemblies vulnerable to pilferage, damage, and loss; to accomplish reduction in cube; and to meet carrier limitations of height, width, and weight. Removed nuts, screws, pins, and washers shall be installed in mating parts and secured to prevent loss. Gaskets and related items shall be individually preserved method IC or III and placed in the toolbox or packaged with other removable parts. Keys shall be secured in keyways of the primary components, attached with shipping documentation, or packaged separately. Disassembly should be limited to parts and components easily removed and installed using no special tools or skilled personnel. Disassembly required to preserve equipment components shall be the minimum necessary to perform preservation.

5.1.1.2 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.3 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservations specified herein are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

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5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces, shall be coated with type P-6 or P-11 preservative and wrapped or covered as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.6 Gears.

5.1.1.6.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or a thin film of primer conforming to TT-P-664 or MIL-P-53030.

5.1.1.6.2 Enclosed gears. Enclosed gears which operate in lubricants shall be preserved by having the housing filled to the operating level with one of the following applicable preservatives:

Operating Lubricant	Preservative Lubricant
Lubricating oil (SAE 10, 30, or 50)	P-10, grade as applicable
Gear lubricant (SAE 75, 80, 85, 90, or 140)	MIL-L-2105, grade as applicable
Lubricant other than above	Manufacturer's approved operating lubricant

Gear mechanism shall be operated to insure coating of all interior surfaces with preservative. Gear housings containing any of the above lubricants shall be identified with a weatherproof tag to indicate: "This housing is filled to the operating level with lubricant. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.7 Drive chains.

5.1.1.7.1 Enclosed chains. Enclosed chains and chain housings shall be preserved and tagged as specified in 5.1.1.6.2 for enclosed gears.

5.1.1.7.2 Exposed chains. Exposed chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chain and sprockets shall be coated with type P-1 preservative.

5.1.1.8 Brakes.

5.1.1.8.1 Brake drum assemblies. All metal surfaces enclosed within brake drums and the braking surface or face of the brake drums shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-53030. Care shall be exercised to prevent the primer from coming in contact with brake lining, rubber, and rubber impregnated parts.

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5.1.1.8.2 Air actuated brake systems. The air actuated brake systems shall be drained of condensation. Interior surfaces of the air supply tanks shall be fogged with type P-10, grade 30 preservative. Excess preservative oil shall be drained. Threaded openings and the threads of drain plugs shall be coated with type P-10. The drain plugs shall be reinstalled. Air line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable conforming to MIL-C-5501 or by sealing the ports with waterproof tape. Attach a waterproof tag to the valves and service lines indicating, "Remove (plugs, caps, or tape) from exhaust ports and valves before operating." Markings shall be applied with a waterproof material.

5.1.1.9 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with hydraulic fluid required for operation. Disconnect the line from the high-pressure outlet of control valve and connect a temporary line to the valve. Disconnect the lower end of the oil return line tube and connect the other end of the temporary line. Remove the air bleed screw from the top of the cylinder. With the engine idling, slowly move the control valve lever to up position allowing hydraulic fluid from reservoir to completely fill the cylinder above the piston. Remove the temporary line and reconnect the high-pressure line and oil return line in their proper position. Reinstall the air bleed screw. For systems not provided with an air bleed screw (e.g., hoses in place of air bleed screws), the void above the piston should be filled in any practical manner which will give in effect the same result. Preserve exposed machined surfaces of operating valve controls and pistons as specified in 5.1.1.5.2. The hydraulic control valves shall be secured in the neutral position. A weatherproof tag shall be attached to the control level indicating: "Drain oil from upper part of lift cylinder by slowly raising the upright assembly so that the cylinder piston is in the top most position. This will allow oil to return to the oil reservoir by oil return line." Markings shall be applied to the tag with a waterproof material.

5.1.1.10 Conveyor belts. Metal belt fasteners shall be coated with water resisting varnish conforming to TT-V-119 or MIL-V-13811. Each belt shall be completely rolled with a tube or piece of round wood placed in each end fold of the belt to prevent sharp bends. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required to hold belt in compact roll, strapping conforming to QQ-S-781, class 1, type I, finish B, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping shall not be tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-1, or E-2, with all laps, seams, and folds sealed with a waterproof adhesive or tape.

5.1.1.11 V-belts, flat belts, and pulleys. Unpainted surfaces of pulleys shall be coated with primer conforming to TT-P-664 or MIL-P-53030. The belts for each unit shall be preserved method IC-1 or IC-3. Large flat belts shall be preserved and packaged as specified for conveyor belts.

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5.1.1.12 Pumps and burners. The interior surfaces of the pump, piping, valves, jets, and nozzles and burners shall be coated with type P-10 preservative in a manner to insure thorough coating of all interior surfaces. Excess preservative shall be drained. All openings shall be sealed with metal or plastic caps or plugs, or sealed with waterproof tape.

5.1.1.13 Refractory linings. Installed refractory linings shall be blocked, braced, and cushioned to prevent breaking or loosening due to shock and vibration during shipment and handling.

5.1.1.14 Fuel tanks. The interior surfaces of fuel tanks, if uncoated or not fabricated or corrosion-resistant material, shall be fogged with type P-10 preservative. Excess preservative shall be drained and the threads of drain plugs shall be coated with type P-10 before installing.

5.1.1.15 Electrical components. Electrical motors shall be preserved level A by the alternate requirements of MIL-E-16298. Other electrical components shall be preserved level A in accordance with MIL-E-17555.

5.1.1.16 Air compressors. The air compressors, parts, and accessories shall be preserved in accordance with MIL-C-3600, level A.

5.1.1.17 Wire rope. The wire rope shall be coated with a preservative conforming to VV-L-751, grade 2 or 3 and secured to prevent movement.

5.1.1.18 Openings. Openings in the equipment which will trap free falling water shall be sealed by any suitable means to prevent the entry of water.

5.1.1.19 Piping equipment. Loose piping equipment required to complete the water-supply system shall be preserved method I or III as applicable. Exposed threaded surfaces shall be protected against damage by the use of metal or plastic thread protectors.

5.1.1.20 Lubrication. Lubricate all grease fittings with type P-11 preservative to include universal joints, chassis suspension and steering linkage, fifth wheels, propeller shaft, and parking brake cable. Wheel bearings requiring a NLGI number 2 grade grease shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.21 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 psi above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to 66.6 percent of the specified operating pressure.

5.1.1.22 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

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5.1.1.23 Consolidation. The dash compartment and tool boxes shall be utilized for the consolidation of loose components, tools, and publications. Tool boxes shall be secured to prevent pilferage. The remainder of the components that will not fit in the dash compartment or tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186. Boxes shall be placed in the cab or secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Commercial. The complete asphalt plant and generator set shall be preserved and packaged in accordance with ASTM D 3951. The asphalt plant and generator set shall be lubricated for operational service.

5.2 Packing. The packing shall be levels A and B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Trailer mounted units (except wheel mounted conveyors). Trailer mounted units shall be shipped mobile and uncrated. Movable components shall be blocked and braced as needed to prevent movement during shipment.

5.2.1.2 Wheel mounted conveyors. The conveyor shall be disassembled. The conveyor belt and electric motor shall be packed in boxes conforming to PPP-B-601, military overseas type or PPP-B-621, class 2. The wheels and chassis components shall be packed in open type crates conforming to MIL-C-52950, styles A or B, type optional. Sections of the conveyor shall be shipped uncrated or bundled. When bundling is used the sections shall be nested, arranged, and secured with suitable wood blocking and battens, as required, to form compact nonshifting bundles. Strapping shall be 0.035 by 1.25 inch flat steel conforming to QQ-S-781, class 1 type I, finish B. The bottom of the secured load shall be fitted with wooden skids extending the full width of the load to accommodate forks and slings for handling.

5.2.1.3 Components not trailer mounted. Components not trailer mounted shall be packed in accordance with the level A requirements of MIL-STD-794. Containers shall be selected from table II of MIL-STD-794 for level A requirements.

5.2.1.4 Piping and accessories. Piping suitable for bundling shall be bundled in secured lifts. The piping shall be nested, arranged, and secured with bolts or steel straps, or a combination of both, to form compact non-shifting bundles. Suitable wood blocking shall be used, as required, under strapping to eliminate large voids or irregular shaped bundles. Strapping shall be 0.023 by 0.75 inch flat steel strapping conforming to QQ-S-781, class I, type I, finish B. Strapping shall be spaced not greater than 36 inches on center, with end straps not greater than 18 inches from each

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end. Metal edge protectors shall be used under strapping when strapping bears on the pipe. The bundles shall be packed in open type crates conforming to MIL-C-52950, styles A or B, type V. Short pieces of pipe, hose fittings, and accessories shall be packed in close-fitting boxes conforming to PPP-B-621, class 2 or PPP-B-601, military overseas type. The contents in the boxes and crates shall be cushioned, blocked, and braced to prevent movement.

5.2.2 Level B. Level B requirements shall be the same as level A except that components not trailer mounted shall be packed in containers designated for level B from table II of MIL-STD-794.

5.2.3 Commercial. The complete asphalt plant shall be prepared for shipment in accordance with ASTM D 3951. Loose components that are subject to damage during shipment shall be blocked, braced, or otherwise secured to prevent movement and damage during shipment. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When sufficient space is available, packaged and unpackaged components required for each unit or section shall be placed and secured on the unit or sections for which intended in any available space that will not interfere with lifting or mobility of the units. Boxed and crated components shall be shipped at the same time as the mobility units. Packing and containers shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The asphalt plant is intended primarily for use in asphaltic paving of airfield, road, street, parking lot, large floor area surfaces, or other uses.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2).
c. Size required (see 1.2).
d. When a type II, style A, class 1 skid mounted diesel engine driven generator set conforming to MIL-G-19826 is required (see 3.1 and 3.10.4).
e. When electromagnetic interference suppression is required for the generator set (see 3.10.4.1).
f. When a single deck scalping screen is required (see 3.13.1).
g. When the asphalt plant control system is to be other than as specified (see 3.18).

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h. When an automatic mixing control system is to be furnished (see 3.19).
i. Means for lubrication, if other than as specified (see 3.21).
j. When treatment for fungus resistance is required (see 3.22).
k. Color of finish coat, if other than specified (see 3.24).
l. When a first article inspection is required (see 4.2.1).
m. When electromagnetic interference suppression testing is required for the generator set (see 4.4.6).
n. Level of preservation and level of packing (see 5.1 and 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
None	DI-E-5325	Engineering Drawing (Commercial)	None
None	DI-M-4022	Commercial Manual	None
None	UDI-A-23264	Certification Data/Report	None

The above DID were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a first article sample or a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Subject term (key word) listing.

Asphalt mixing plant, portable
Drum type mixer
Trailer mounted

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review Activity:
Army - CE
Air Force - 84

Preparing activity:
Navy - YD

(Project 3895-0291)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-29212B	MILITARY SPECIFICATION - MIXING PLAN ASPHALT, DRUM TYPE, PORTABLE, ELECTRIC MOTOR DRIVEN, 120 TONS PER HOUR CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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